HeartQuest AR

An app that helps students learn about human heart anatomy
through interactive 3D exploration and quizzes.

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(In cele ce urmează, considerați versiunea ideală a aplicației pe care doriți să o implementați. Cât la sută va fi și implementat până la finalul semestrului rămâne de văzut mai târziu.)

User Story Map (2p)

1. **User 1**
   The primary user of my application is a Medical or Biology Student.
   a) **Goal 1. They want/need to explore and understand the spatial anatomy of the human heart.**
      i. **Activity 1. To do this, in our app, they will be able to place and interact with the 3D model in their physical space.**
      1. **Feature 1. AR surface detection and 3D model placement (desk or floor).**
      2. **Feature 2. Touch gestures for 360-degree rotation, panning, and zooming.**
      3. **Feature 3. A simple reset button to quickly return the 3D model to its default size and position.**

      ii. **Activity 2. Then, they will study the specific anatomical structures.**
      1. **Feature 1. Tap-to-select functionality on specific 3D structures.**
      2. **Feature 2. Informational pop-up cards displaying the anatomical name.**
      3. **Feature 3. Extra information about physiological functions.**

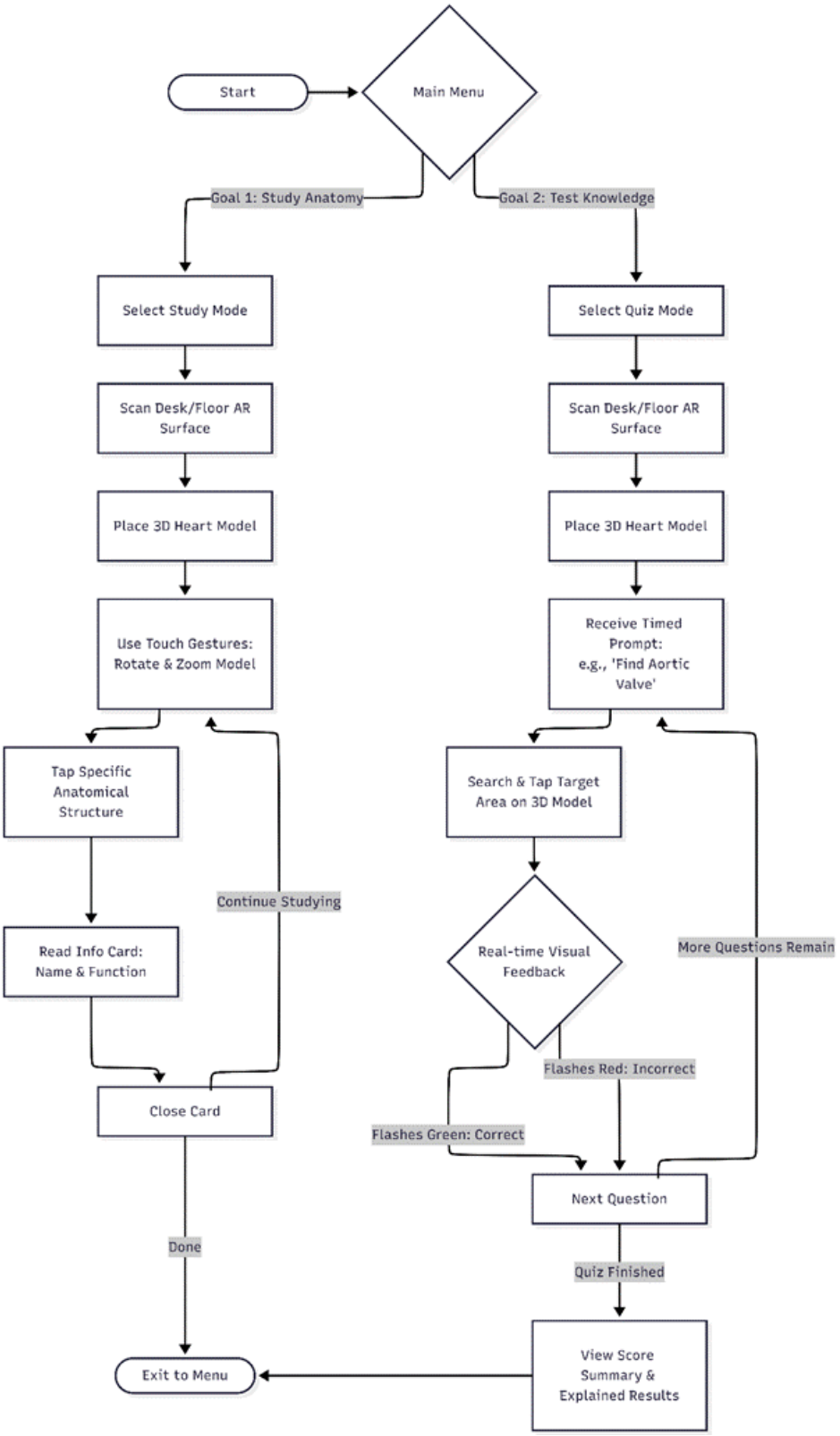
   b) **Goal 2. They want/need to test their knowledge and prepare for an upcoming anatomy exam.**
      i. **Activity 1. To do this, in our app, they will take interactive AR spatial quizzes.**
      1. **Feature 1. Timed location prompts (e.g., the app asks "Find the Aortic Valve").**
      2. **Feature 2. Real-time visual feedback.**
      3. **Feature 3. End-of-quiz score summary with explained results.**

2. **(eventual) User 2**
   The secondary user of my application is an Anatomy teacher.

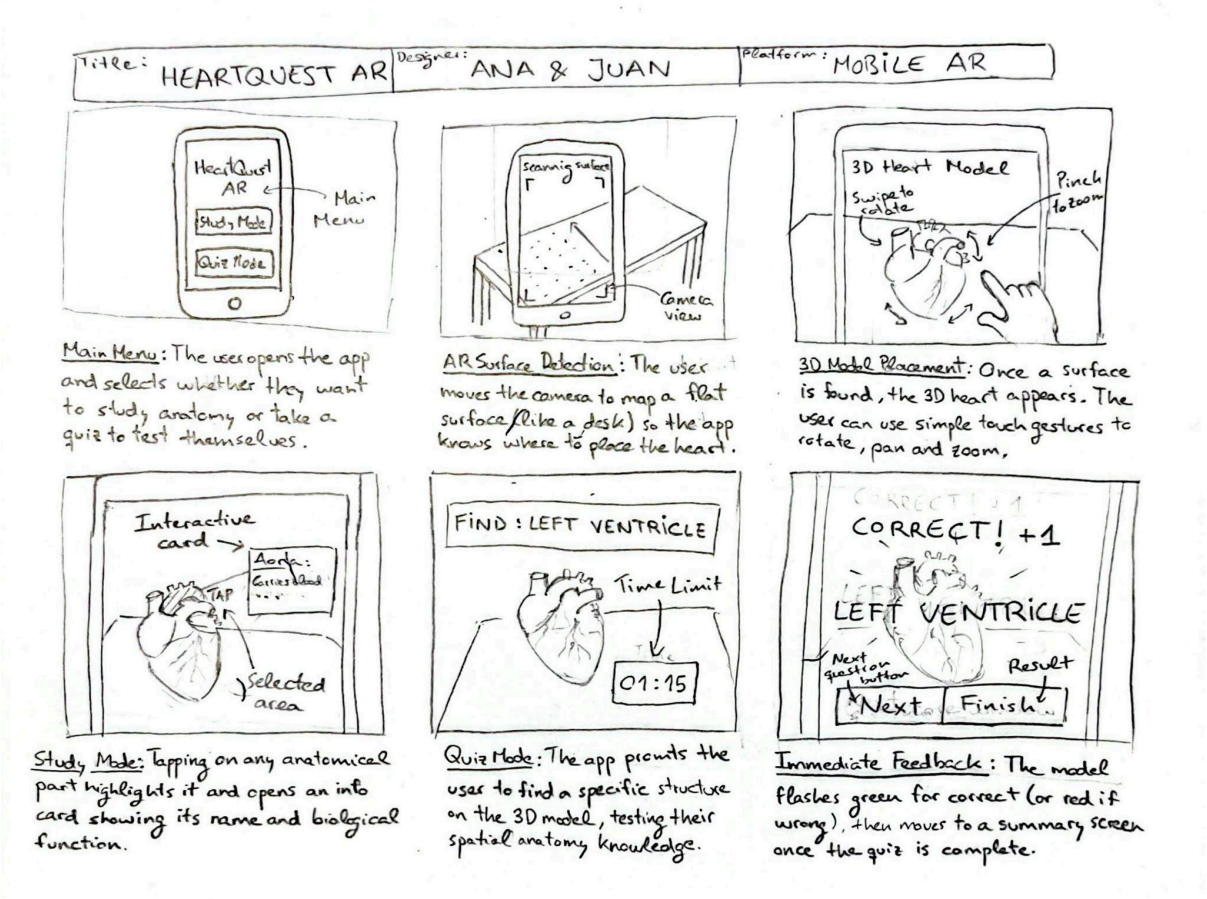
a) **Goal 1. They want/need to assess their students' practical understanding of the heart's structure without relying on physical plastic models.**
    i. **Activity 1. To do this, in our app, they will utilize the application as a fast, interactive examination tool during lab sessions.**
        1. **Feature 1. End-of-quiz score summary, allowing the teacher to instantly check the final grade and identify areas of confusion.**
        2. **Feature 2. Real-time visual feedback, so the instructor can observe the student's decision-making process live as they tap the model.**
        3. **Feature 3. Timed location prompts, ensuring the assessment is standardized and fast-paced for each student.**

b) **Goal 1. They want/need to clearly illustrate complex anatomical relationships to the class during lectures.**
    i. **Activity 1. To do this, in our app, they will manipulate the digital heart in real-time while explaining biological concepts.**
        1. **Feature 1. Tap-to-select functionality, which acts as a digital pointer to bring up specific names and functions on screen for the class to see.**
        2. **Feature 2. 360-degree rotation and scaling (zoom) to easily highlight hidden or smaller internal structures to the audience.**
        3. **Feature 3. AR surface placement, seamlessly turning any flat classroom desk into an engaging 3D presentation stage.**

Journey Map (2p)



Wireframing/Storyboarding (3p)



Planning Asset Creation

3D

- ➔ Realistic Human Heart Model (Downloaded from Sketchfab or another 3D asset library).
- ➔ 2D Anatomical Labels and Selection Indicators (Created using Unity UI).

Graphics & 2D

- ➔ UI Layouts, Information Cards & Menus (Created directly using Unity UI).
- ➔ HeartQuest App Logo & Splash Screen Illustration (AI Generated).
- ➔ UI Icons: Back, Close, Reset and Quiz (Google Material Icons).
- ➔ Basic Touch Gesture Indicators: Rotate and Zoom (Flaticon).

Sounds

- ➔ Optional Background Music for Study Mode (Royalty-free audio).
- ➔ Optional Heartbeat Sound Effect (Royalty-free audio).
- ➔ Optional UI Click and Quiz Feedback Sounds (Kenney UI Audio).

VFX

- ➔ Basic AR Surface Detection and Heart Placement (Unity AR Foundation).
- ➔ Simple Anatomical Structure Selection Feedback (Unity UI Highlight or Color Change).
- ➔ Basic Quiz Feedback (Green/Red UI Messages for Correct and Incorrect Answers).

Other Assets / Libraries

- ➔ AR Foundation Framework for Unity (Used for surface detection, tracking and camera passthrough).
- ➔ Unity Input System (Used for basic touch gestures such as rotation and pinch-to-zoom).
- ➔ Unity UI (Used for menus, buttons, anatomical information cards and quiz interfaces).